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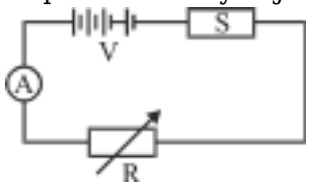
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PHYSICS

1) Figure shows a piece of semiconductor (pure one) S in series with a variable resistor R and a source of constant voltage V. S is heated and the current is kept constant by adjustment of R. Which of the following factors will decrease during this process :-



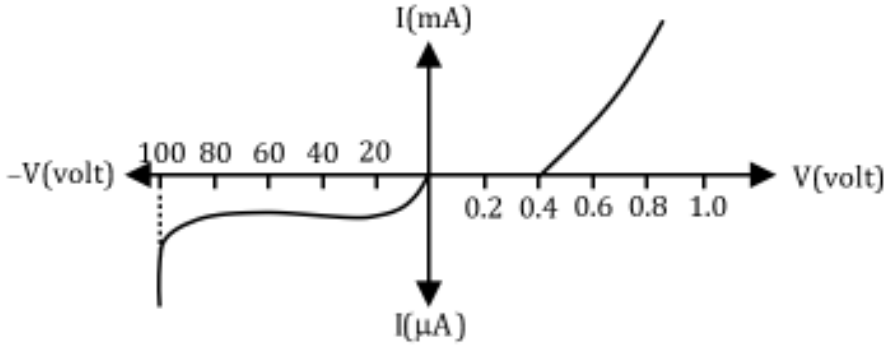
(1) The drift velocity of the conduction electrons in S.
(2) The DC resistance of S
(3) The number of conduction electrons in S.
Which of the following is correct ?

(1) only 1
(2) 1 and 2
(3) 1, 2 and 3
(4) only 3

2) A potential difference of 2V is applied between the opposite faces of a Ge crystal plate of area 1 cm^2 and thickness 0.5 mm. If the concentration of free electrons in Ge is $2 \times 10^{19}/m^3$ and mobilities of electrons and holes are 0.36 $\frac{m^2}{volt-s}$ and 0.14 $\frac{m^2}{volt-s}$ respectively, then the current flowing through the plate will be :-

(1) 0.25 A
(2) 0.45 A
(3) 0.56 A
(4) 0.64 A

3) I-V characteristic of a diode is shown in figure.



If this diode is connected across a battery of emf 10 V and a resistance of 12 kΩ in series, the current through diode in forward bias is nearly :

(1) 0.4 mA

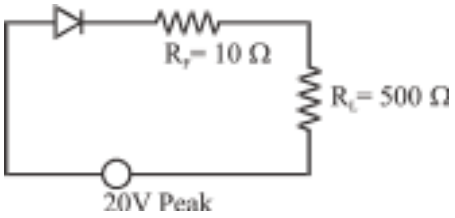
(2) 0.83 mA

(3) 0.8 mA

(4) 1.2 mA

4)

In circuit shown the barrier voltage of diode is 0.7 V. Match the physical quantities :-



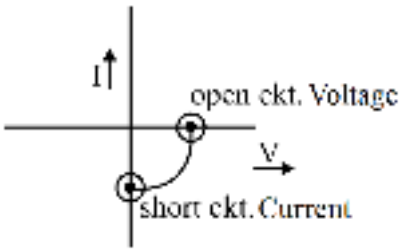
Column-I		**Column-II**	
(A)	Peak current in diode (in mA)	(P)	37.8
(B)	Peak voltage (in volt) at the end of load	(Q)	40.0
(C)	Peak current (in mA), if diode is ideal	(R)	20.0
(D)	Peak voltage (in volt) at the ends of load, if diode is ideal	(S)	18.9

(1) A → Q ; B → R ; C → P ; D → S

(2) A → P ; B → S ; C → R ; D → Q

(3) A → P ; B → S ; C → Q ; D → R

(4) A → S ; B → P ; C → Q ; D → R



5) Given characteristic curve is related with Electronic device :

(1) Photo diode

(2) Zener diode

(3) Solar cell

(4) Light emitting diode (LED)

6) Which of the following Boolean expression is not correct ?

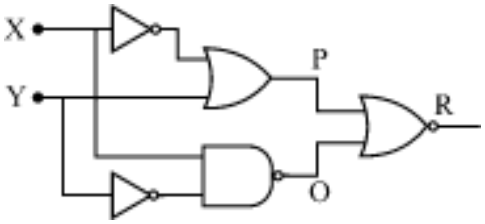
(1) $B + A\bar{B} = A + B$

(2) $A \cdot (\bar{A} + B) = A \cdot B$

(3) $A + B \cdot C = A \cdot B + A \cdot C$

(4) $(\bar{A} + B) \cdot (A + C) = \bar{A}.C + A.B + B.C$

7) Figure gives a system of logic gates. From the study of truth table it can be found that to produce a high output (1) at R, we must have :-



(1) X = 0, Y = 1

(2) X = 1, Y = 1

(3) X = 1, Y = 0

(4) X = 0, Y = 0

8) ***Assertion* (A) :-** A P-N photodiode is made from a semiconductor for which E_g = 2.8 eV. This photo diode will not detect the wavelength 600 nm. ***Reason* (R) :-** A PN photodiode detect wavelength λ, if $\frac{hc}{\lambda} > E_g$.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

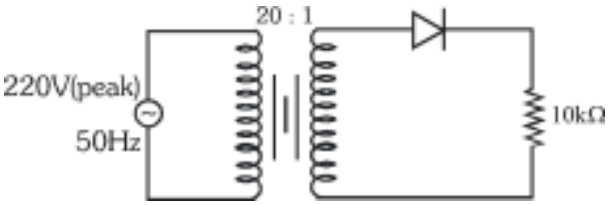
(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

9)

In given circuit, determine value of RMS current through 10 kΩ resistance, used p-n junction is ideal.



(1) 0.55 mA

(2) 5.5 mA

(3) 0.77 mA

(4) 1.1 mA

10) A ray of light parallel to principle axis is incident on a concave mirror at angle of incidence 37°. Reflected ray meets the principle axis at point B. Distance of B from pole of mirror is :-

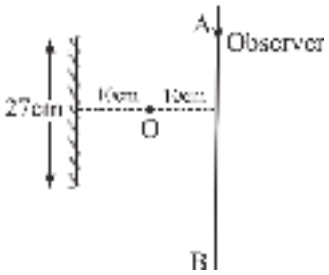
(1) $\frac{3R}{8}$

(2) $\frac{5R}{8}$

(3) $\frac{R}{2}$

(4) $\frac{3R}{4}$

11) An observer is moving along the line AB as shown. When image of object O is first visible to observer, he starts from rest with the acceleration of $2cm/s^2$, then the time for which image is visible



to observer is :

(1) 3 sec

(2) 9 sec

(3) 12 sec

(4) None of the above

12) For real object match the corresponding entries of column-1 with column-2 (where m is the magnification produced by the mirror) :-

	Column-1		**Column-2**
A	m = –4	a	Convex mirror
B	m = –1/4	b	Concave mirror
C	m = +3	c	Real image
D	m = +1/3	d	Virtual image

(1) A-a and c, B-a and d, C-a and b, D-c and d

(2) A-a and d, B-b and c, C-b and d, D-b and d

(3) A-c and d, B-b and d, C-b and c, D-a and d

(4) A-b and c, B-b and c, C-b and d, D-a and d

13) The nature of image of a candle flame located 40 cm from a concave spherical mirror is real, inverted and magnified four times. Then the radius of curvature of the mirror is :

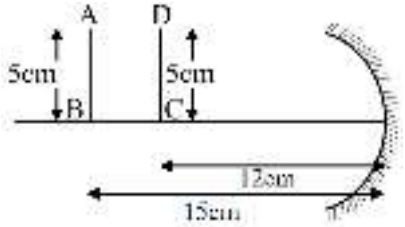
(1) 32 cm

(2) 64 cm

(3) 48 cm

(4) 80 cm

14) A U-shaped wire is placed in front of radius of curvature 20 cm as shown in figure. The total length of image of wire ABCD will be :-

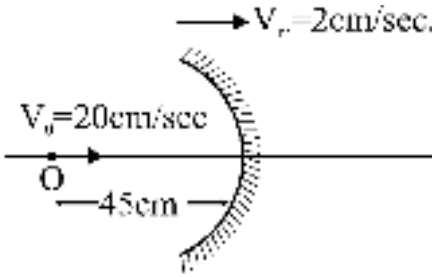


(1) 65 cm

(2) 50 cm

(3) 80 cm

(4) 75 cm



15) Velocity of image of object O will be :-

(1) 36 cm/sec

(2) 70 cm/sec

(3) 80 cm/sec

(4) 74 cm/sec

16) The thickness of a slab is $5\sqrt{3}$ cm and refractive index $\sqrt{3}$. If light incident on it at angle of incidence 60°, then find lateral shift between direction of incident ray and emergent ray produced by it :-

(1) $\frac{5\sqrt{3}}{2}$cm

(2) $\frac{5}{2}$cm

(3) $\frac{5}{\sqrt{3}}$cm

(4) 5 cm

17) An object is put at a distance of 5 cm from the first focus of a convex lens of focal length 10cm.If a real image is formed, its distance from the lens will be :–

(1) 15 cm

(2) 20 cm

(3) 25 cm

(4) 30 cm

18) Two thin lenses of power +15D and –5D are placed coaxially in contact with each other. The position of image of an object placed at 20 cm from the combination will be :

(1) +10 cm

(2) –10 cm

(3) +20 cm

(4) –20 cm

19) An object is placed at 15 cm from a convex lens of focal length 10 cm. Where should a convex mirror of radius 12 cm be placed from lens such that image will coincide with object?

(1) 18 cm

(2) 17 cm

(3) 14 cm

(4) 20 cm

20) Two thin lens have a combined power of 10 D in contact. When separated by 20 cm, their equivalent power is 6.25 D. Find their individual powers in dioptres :

(1) 3.5 and 6.5

(2) 5 and 5

(3) 7.5 and 2.5

(4) 9 and 1

21) ***Assertion*** :- In a simple microscope, image formed is virtual, erect and magnified.
Reason :- Object to be seen is held between optical centre and focus of the convex lens.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

22) The focal length of objective and eye lens of a microscope are 4 cm and 8 cm respectively. If the least distance of distinct vision is 24 cm and object distance is 4.5 cm from the objective lens, then the magnifying power of the microscope will be (Final image is at infinity) :-

(1) 18

(2) 32

(3) 24

(4) 20

23) A compound microscope consists of an objective lens of focal length 2 cm and an eye piece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at the least distance of distinct vision (25 cm)?

(1) 2.5 cm

(2) 1.67 cm

(3) 2 cm

(4) 3.3 cm

24) In Gallilean telescope, if the powers of an objective and eye lens are respectively +1.25 D and –20 D; then for relaxed vision, the length and magnification will be :

(1) 21.25 cm and 16

(2) 75 cm and 20

(3) 75 cm and 16

(4) 8.5 cm and 21.25

25) When an object is kept at a distance of 5 cm from the pole of a concave mirror, a virtual image is formed. If the object distance is 15 cm, a real image is formed. Both the images have same magnification. The focal length of the mirror is

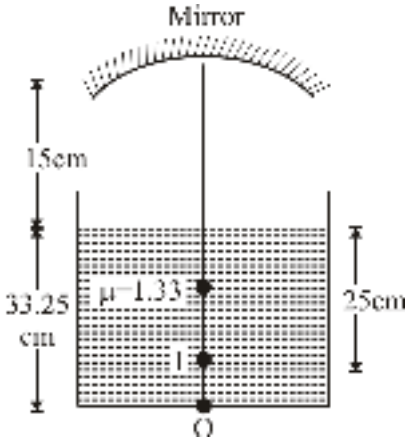
(1) –10 cm

(2) –5 cm

(3) –20 cm

(4) –15 cm

26) A container is filled with water (μ = 1.33) upto a height of 33.25 cm. A concave mirror is placed 15 cm above the water level and the image of an object placed at the bottom of container is formed 25 cm below the water level. The focal length of the mirror is



(1) 10 cm

(2) 15 cm

(3) –18.3 cm

(4) 25 cm

27) Magnification produced by astronomical telescope for normal adjustment is 20 and length of telescope is 2.1 m. The magnification, when the image is formed at least distance of distinct vision (D = 25 cm), is -

(1) 14

(2) 6

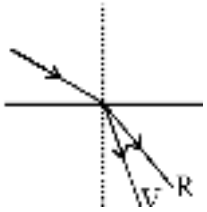
(3) 16

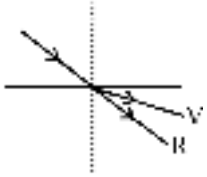
(4) 28

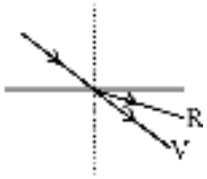
28) A ray of light is coming from air to water. Which of the following figures show dispersion of light?

(1)



(2) 

(3) 

(4) 

29) A light bulb is lit at the bottom of tank filled with liquid ($\mu = 2$) upto a height of 3m. The total surface area (in m^2) which is illuminated by light is :-

(1) π

(2) 2π

(3) 3π

(4) 4π

30) The focal length of a convex lens of glass ($\mu = 1.5$) is 2 cm. The focal length of the lens, when immered in a liquid of refractive index 1.25, will be :-

(1) 5 cm

(2) 2.4 cm

(3) 1 cm

(4) 4 cm

31) In a laboratory four convex lenses L_1, L_2, L_3 and L_4 of focal lengths 2, 4, 6 and 8 cm, respectively are available. Two of these lenses form a telescope of length 10 cm and magnifying power 4. The objective and eye lenses are respectively

(1) L_2, L_3

(2) L_2, L_4

(3) L_1, L_2

(4) L_4, L_1

32) Magnification of a compound microscope is 30. Focal length of eye–piece is 5cm and the final image is formed at least distance of distinct vision. Magnification of objective lens is :-

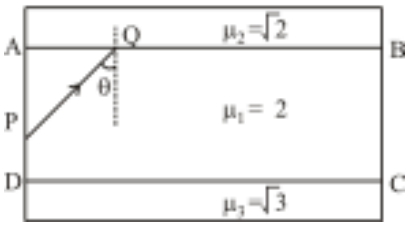
(1) 6

(2) 5

(3) 7.5

(4) 10

33) A parallel sides slab ABCD of refractive index 2 is sandwiched between two slabs of refractive indices $\sqrt{2}$ and $\sqrt{3}$, as shown in the figure. The minimum value of angle θ such that the ray PQ suffers total internal reflection at both the surface AB and CD is

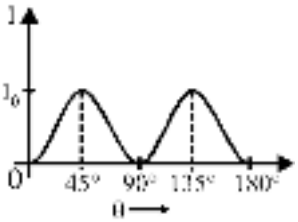


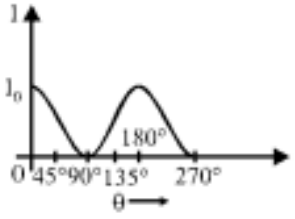
(1) 30°
(2) 45°
(3) 60°
(4) 75°

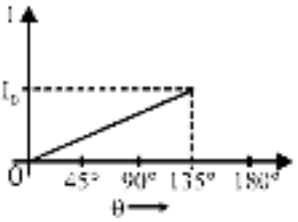
34) For a given medium, the polarizing angle is 60°. What will be the critical angle for this medium?

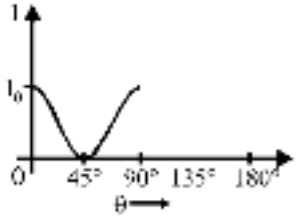
(1) $\sin^{-1}$ (0.577)
(2) $\sin^{-1}$ (0.177)
(3) $\sin^{-1}$ (0.277)
(4) $\sin^{-1}$ (0.415)

35) The graph showing the dependence of intensity of transmitted light on the angle between polariser and analyser is (I_0 is intensity of plane polarised light)

(1)


(2)


(3)


(4) 

36) A beam of natural light falls on a system of 5 polaroids, which are arranged in succession such that the pass axis of each polaroid is turned through 60° with respect to the preceding one. The fraction of the intensity of incident light that passes through the system is :

(1) $\frac{1}{64}$

(2) $\frac{1}{32}$

(3) $\frac{1}{256}$

(4) $\frac{1}{512}$

37) The fringe width in the Young's double slit experiment is 2×10^{-4} m. If the distance between the slits is halved and the slit screen distance is double, then the new fringe width will be:-

(1) 2×10^{-4} m

(2) 1×10^{-4} m

(3) 0.5×10^{-4} m

(4) 8×10^{-4} m

38) In a Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment -

(1) 200nm

(2) 600nm

(3) 400nm

(4) None of these

39) The central fringe of the interference pattern produced by the light of wavelength 6000 Å is found to shift to the position of 4th dark fringe, after a glass sheet of refractive index 1.5 is introduced. The thickness of glass sheet would be :-

(1) 4.8 μm

(2) 4.2 μm

(3) 5.4 μm

(4) 3.0 μm

40) In Young's double slit experiment the y-co-ordinates of central maximum and 10th maxima are 2 cm and 5 cm respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5, the corresponding y-co-ordinates will be :

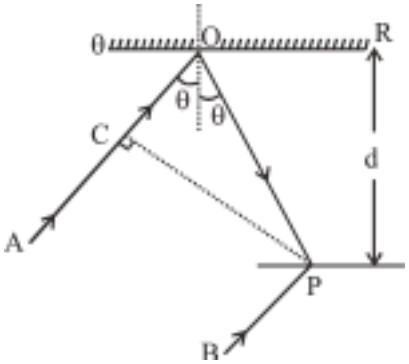
(1) 2 cm, 7.5 cm

(2) 3 cm, 6 cm

(3) 2 cm, 4 cm

(4) $\frac{4}{3}$cm, $\frac{10}{3}$cm

41) In the diagram, CP represents a wavefront and AO and BP, the corresponding two rays. Find the condition on θ for constructive interference at P between the ray BP and reflected ray OP.



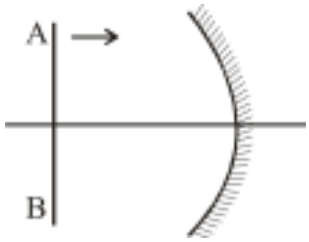
(1) $\cos\theta = \frac{3\lambda}{2d}$

(2) $\cos\theta = \frac{\lambda}{4d}$

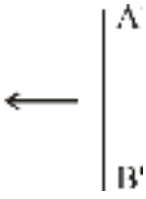
(3) $\sec\theta - \cos\theta = \frac{\lambda}{d}$

(4) $\sec\theta - \cos\theta = \frac{4\lambda}{d}$

42) A plane wavefront AB is incident on a concave mirror as shown. Then the reflected wavefront just after reflection from mirror will be :-



(1)



(2)



(3) 

(4) None of above

43) In Young's double slit experiment, the wavelength of red light is 7800 Å and that of blue light is 5200 Å. The value of n for which n^{th} bright band due to red light coincides with $(n + 1)^{th}$ bright band due to blue light, is :-

(1) 1

(2) 2

(3) 3

(4) 4

44) A plate of thickness t made of a material of refractive index μ is placed in front of one of the slits in a double slit experiment. What should be the minimum thickness t for which the intensity at the centre of the fringe pattern is zero :-

(1) $\frac{(\mu-1)\lambda}{2}$

(2) $(\mu-1)\lambda$

(3) $\frac{\lambda}{2(\mu-1)}$

(4) $\frac{\lambda}{(\mu-1)}$

45) Two coherent light sources having intensity in the ratio 2x produce an interference pattern. The ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be :

(1) $\frac{2\sqrt{2x}}{x+1}$

(2) $\frac{\sqrt{2x}}{2x+1}$

(3) $\frac{\sqrt{2x}}{x+1}$

(4) $\frac{2\sqrt{2x}}{2x+1}$

CHEMISTRY

1) Match the Column-I with Column-II

	Column-I (Salt)		**Column-II (pH at 25°C)**

(A)	K_2S	(p)	pH = 7
(B)	NaCl	(q)	pH < 7
(C)	NH_4Cl	(r)	pH may be less than 7 or more than 7 depending on K_a and K_b
(D)	Salt of weak acid & weak base	(s)	pH > 7

(1) (A) → (p); (B) → (q); (C) → (r); (D) → (s)

(2) (A) → (s); (B) → (p); (C) → (q); (D) → (r)

(3) (A) → (s); (B) → (r); (C) → (q); (D) → (p)

(4) (A) → (p); (B) → (s); (C) → (r); (D) → (q)

2) **Statement-1 :** On dilution of a concentrated solution of CH_3COOH , the concentration of $[H^+]$ decreases.
Statement-2 : Increase in volume is more than the increase in degree of ionisation.

(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true.

3) Conjugate base for Bronsted acids H_2O and HF are :-

(1) OH^- and H_2F^+ respectively

(2) H_3O^+ and F^-, respectively

(3) OH^- and F^-, respectively

(4) H_3O^+ and H_2F^+, respectively

4) Which of the following statement(s) is/are correct.
(A) The pH of 1×10^{-8} M HCl is 8
(B) The conjugate base of $H_2PO_4^-$ is HPO_4^{2-}
(C) k_w increases with increase in temperature
(D) When a solution of weak monoprotic acid is titrated against a strong base at half neutralisation point
$\Rightarrow$ pH = $\frac{1}{2}$ pKa.

(1) A, D

(2) B, C

(3) A, B, C

(4) B, C, D

5) 0.1M CH_3COOH is titrated against 0.1M NaOH. What would be the difference in pH between 1/4 and 3/4 stages of neutralisation of acid :-

(1) $2\log\frac{3}{4}$

(2) $2\log\frac{1}{4}$

(3) $\log\frac{8}{3}$

(4) 2log 3

6)

Consider the following :
a. Sodium chloride
b. Sodium nitrate
c. Silver chromate (Ag_2CrO_4)

Which is the correct increasing order of the solubility of silver chloride in 0.1 M solution of the above compounds ?

(1) a < b < c

(2) b < c < a

(3) c < a < b

(4) a < c < b

7) The degree of hydrolysis of 0.1 M $R\overset{+}{N}H_3\overset{-}{Cl}$ solution is 1.0%. If the concentration of $R\overset{+}{N}H_3\overset{-}{Cl}$ is made 0.4 M, what is the new degree of hydrolysis (in percentage) ?

(1) 0.01

(2) 0.001

(3) 0.2

(4) 0.5

8) K_a for CH_3COOH is 1.8×10^{-5}. Find out the percentage dissociation of 0.2M CH_3COOH in 0.1M HCl solution

(1) 0.018 %

(2) 0.36 %

(3) 18 %

(4) 36 %

9) The solubility of salt in saturated aqueous solution of NH_4KHPO_3 is:-

(1) $(K_{sp})^{1/3}$

(2) $\left(\frac{K_{sp}}{4}\right)^{1/3}$

(3) $(K_{sp})^{1/4}$

(4) $\left(\frac{K_{sp}}{4}\right)^{1/4}$

10) Which is correct for acidic solution :-

(1) $[H^+] > [OH^-]$
(2) pOH > 7
(3) $[OH^-] < 10^{-7}$
(4) All of these

11) Most disordered system is :-

(1) 1 mol of O_2 at NTP
(2) 1 mol fo N_2 at NTP
(3) 1 mol of air at NTP
(4) 1 mol of CO_2 at NTP

12) Which of the following pair is an example of extensive property :-

(1) P and T
(2) C_{pm} and E
(3) E and V
(4) T and V

13) When two moles of hydrogen expands isothermally against a constant pressure of 1 atm, at 25°C from 15 L to 50 L, the work done (in litre-atm) will be:

(1) –17.5
(2) –35
(3) –51.5
(4) –70

14) The value of enthalpy change (ΔH) for the reaction,
$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(I)$ at 27°C is –1366.5 kJ mol^{-1}. The value of internal energy change for the above reaction at this temperature will be

(1) –1369.0 kJ
(2) –1364.0 kJ
(3) –1361.5 kJ
(4) –1371.5 kJ

15) For the equilibrium $H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$ at 1 atm and 298K :

(1) Standard free energy change is equal to zero ($\Delta G^\circ = 0$)
(2) Free energy change is less than zero ($\Delta G < 0$)
(3) Standard entropy change is less than zero ($\Delta S^\circ < 0$)
(4) Standard free energy change is greater than zero ($\Delta G^\circ > 0$)

16) Match the following columns:-

	Column-I		Column-II
i	In adiabatic process	a	$\Delta H = q$
ii	In isochoric process	b	$\Delta E = w$
iii	In isothermal process	c	$\Delta E = q$
iv	In isobaric process	d	$q = -w$

(1) (i) - (b), (ii) - (c), (iii) - (d), (iv) - (a)

(2) (i) - (c), (ii) - (b), (iii) - (d), (iv) - (a)

(3) (i) - (b), (ii) - (c), (iii) - (a), (iv) - (d)

(4) (i) - (a), (ii) - (c), (iii) - (d), (iv) - (b)

17) If x_1, x_2 and x_3 are enthalpies of H–H, O=O and O–H bonds respectively, and x_4 is the enthalpy of vaporisation of water, estimate the standard enthalpy of combustion of hydrogen.

(1) $x_1+\frac{x_2}{2}-2x_3+x_4$

(2) $x_1+\frac{x_2}{2}-2x_3-x_4$

(3) $x_1+\frac{x_2}{2}-x_3+x_4$

(4) $2x_3-x_1-\frac{x_2}{2}-x_4$

18) Consider following reactions :-

(a) $N_2(g) + O_2(g) \rightarrow 2NO(g)$; $\Delta H = X_1$

(b) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$; $\Delta H = X_2$

(c) $2H_2O(g) + O_2(g) \rightarrow 2H_2O_2(g)$; $\Delta H = X_3$

(d) $PCl_3(g) + Cl_2(g) \rightarrow PCl_5(g)$; $\Delta H = X_4$

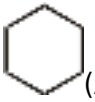
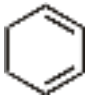
In which case(s), $\Delta H_f = \frac{\Delta H}{2}$

(1) a, b, c

(2) d

(3) b, c

(4) a

19) If  $+ H_2 \rightarrow$  ($\Delta H = -x$ KJ), then amount of heat released for reaction  $+ 2H_2 \rightarrow$  will be :-

(1) 2x

(2) x

(3) > 2x

(4) < 2x

20) Calculate P–Cl bond enthalpy
Given : $\Delta_f H(PCl_3,g)$ = 306 kJ/mol ;
$\Delta H_{atomization}$ (P,s) = 314 kJ/mol ;
$\Delta_f H(Cl,g)$ = 121 kJ/mol

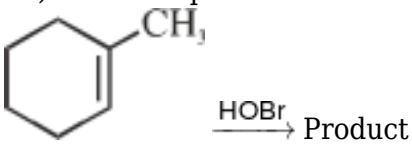
(1) 123.66 kJ/mol

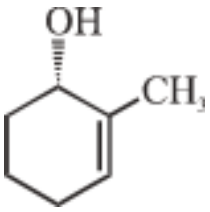
(2) 371 kJ/mol

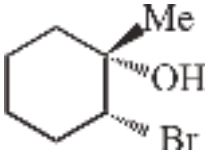
(3) 19 kJ/mol

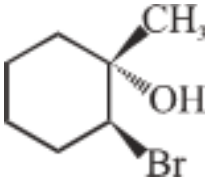
(4) 23 kJ/mol

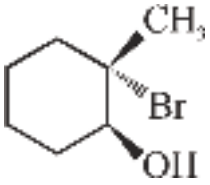
21) The main product of following reaction will be :

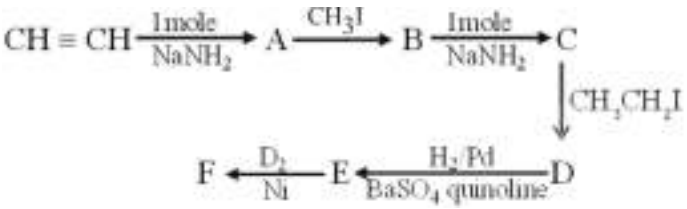


(1) 

(2) 

(3) 

(4) 

22) 

Incorrect statement about the conversions given :-

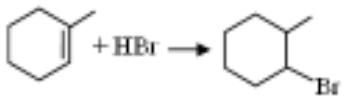
(1) D to E conversion is a syn-addition of H_2

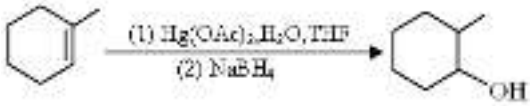
(2) F is a meso compound and is optically inactive

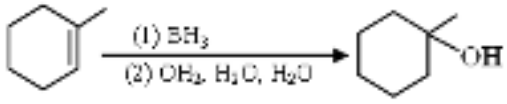
(3) B can gives red precipitate with ammonical cuprous chloride

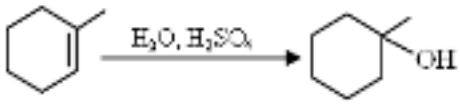
(4) C to D conversion is a nucleophilic substitution and follows SN^2 mechanism

23) Which of the following reactions generates the correct major product ?

(1) 

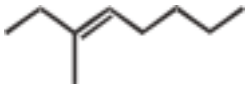
(2) 

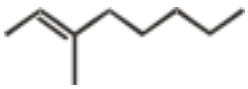
(3) 

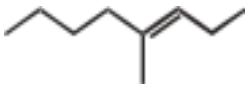
(4) 

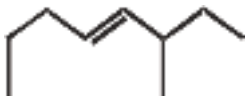
24) Alkene (X) 

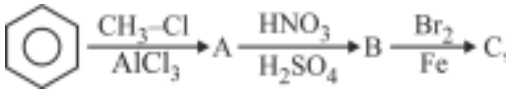
X is :-

(1) 

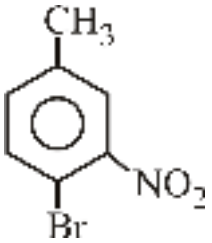
(2) 

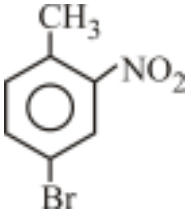
(3) 

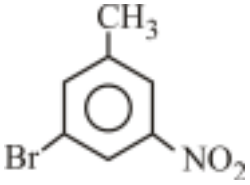
(4) 

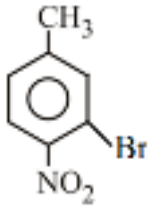
25) 

C is :-

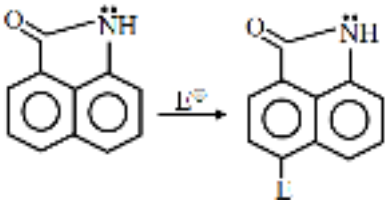
(1) 

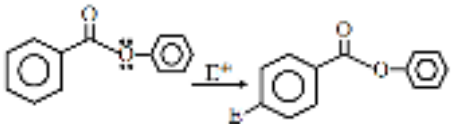
(2) 

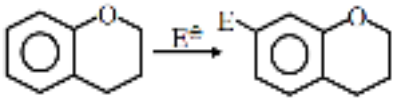
(3) 

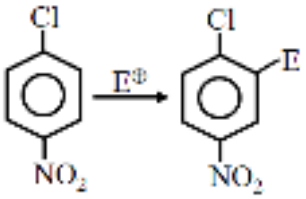
(4) 

26) In which chemical reaction $E^{\oplus}$ attack is correct-

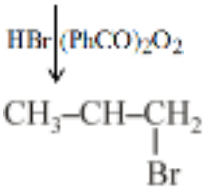
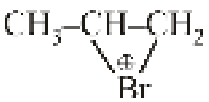
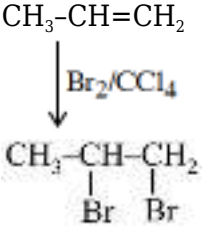
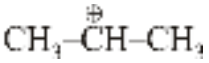
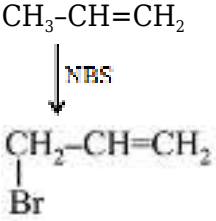
(1) 

(2) 

(3) 

(4) 

27) Match the following column :

Column-I (Reaction)		Column-II (Intermediate)	
(A)	$CH_3–CH=CH_2$ 	(P)	
(B)	$CH_3–CH=CH_2$ 	(Q)	
(C)	$CH_3–CH=CH_2$ 	(R)	

(D)	$CH_3-CH=CH_2$ $\downarrow HBr$ $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$	(S)	$CH_3-\overset{\bullet}{C}H-\underset{\underset{Br}{|}}{CH_2}$

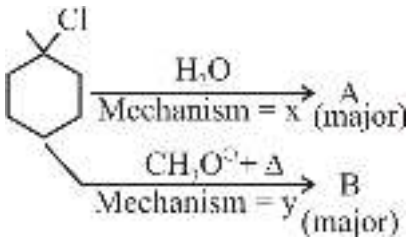
(1) (A) - S, (B) - P, (C) - R, (D) - Q

(2) (A) - Q, (B) - P, (C) - R, (D) - S

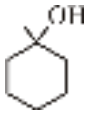
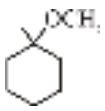
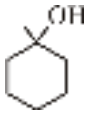
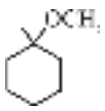
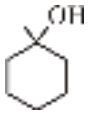
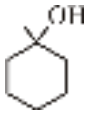
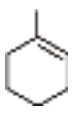
(3) (A) - S, (B) - R, (C) - P, (D) - Q

(4) (A) - S, (B) - P, (C) - Q, (D) - R

28)



Choose correct option for mechanism x and y, product A and B :-

	x	y	A	B
(1)	S_N^2	S_N^2		
(2)	S_N^1	S_N^2		
(3)	S_N^1	E_2		
(4)	S_N^1	E_1		

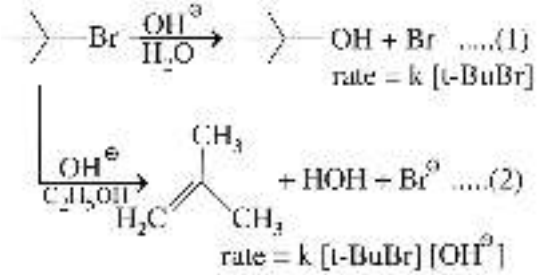
(1) 1

(2) 2

(3) 3

(4) 4

29) Consider the reaction sequence given below :



Which of the following statements is true :

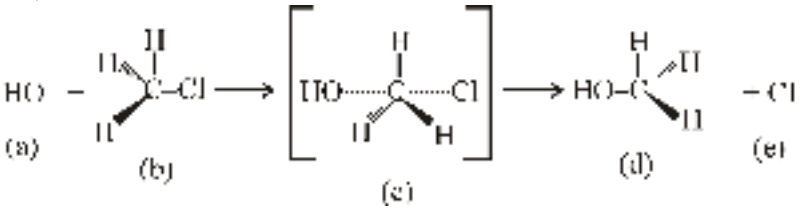
(1) Changing the concentration of base will have no effect on reaction (1)

(2) Changing the concentration of base will have no effect on reaction (2)

(3) Changing the concentration of base will effect reaction (1)

(4) Doubling the concentration of base will double the rate of both the reactions.

30) Which of the statements is incorrect about below reaction ?



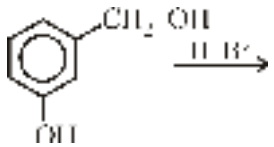
(1) The given reaction follows S_N2 mechanism

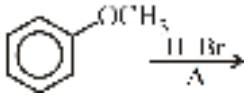
(2) In (c) carbon atom is sp^3 hybridised

(3) In (c) carbon atom is sp^2 hybridised

(4) This reaction follows second order kinetics

31) In which reaction halide is not obtained ?

(1) 

(2) 

(3) $CH_3–CH_2–OH + SOCl_2 \rightarrow$

(4) $CH_3–CH_2–OH + KBr \rightarrow$

32) **Statement-I :** In Aryl halide if electron withdrawing groups are present at ortho and para positions then rate of nucleophilic substitution reaction increases.
Statement-II : In nucleophilic substitution reaction of aryl halides, intermediate carbocation is formed.

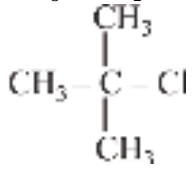
(1) Both statement -I and statement -II are correct

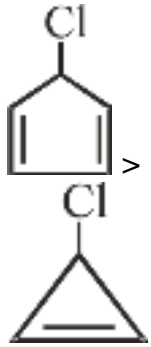
(2) Both statement - I and statement -II are incorrect.

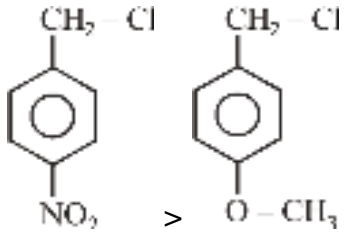
(3) Statement - I is correct but II is incorrect.

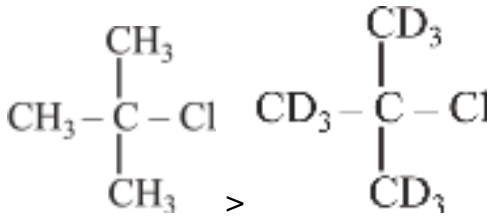
(4) Statement - I is incorrect but II is correct.

33) Which one is correct for rate of hydrolysis :-

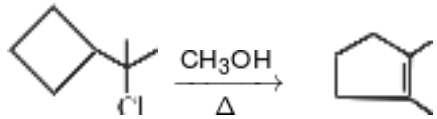
(1) $CH_3 – CH_2 – Cl >$ 

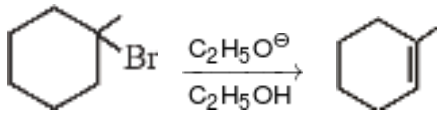
(2)  >

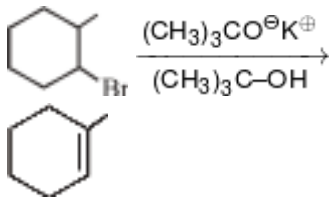
(3) 

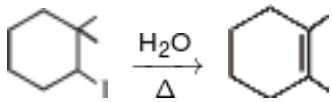
(4) 

34) Which of the following reaction does not represent major product ?

(1) 

(2) 

(3) 

(4) 

35) Arrange the following in correct order of nucleophilicity :-

(I) CH_3–O–CH_2–$O^{\ominus}$ (II) Ph–CH_2–$O^{\ominus}$

(III) CH_3–$O^{\ominus}$ (IV) CH_3CH_2–$S^{\ominus}$

(1) IV > III > II > I

(2) I > II > III > IV

(3) III > IV > II > I

(4) II > I > IV > III

36) Select the correct option for matching of column-I with column-II

	Column-I		Column-II
(A)	Benzene + Toluene	(P)	Non-ideal solution
(B)	Ethanol + Water	(Q)	Ideal solution
(C)	Benzene + Ethanol	(R)	$\Delta H_{mix} > 0$
(D)	Acetone + Chloroform	(S)	$\Delta V_{mix} < 0$

(1) (A) → P; (B) → Q, R; (C) → P, R; (D) → S

(2) (A) → Q; (B) → P, R; (C) → P, R; (D) → P, S

(3) (A) → Q; (B) → P, S; (C) → P, R; (D) → P, R

(4) (A) → Q; (B) → P, R; (C) → P, S; (D) → P, R

37) **Statement–I** :- K_H (Henry's law constant) has the same unit as that of pressure
Statement–II :- Different gases have different values of K_H at same temperature

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct and Statement II is incorrect.

(3) Statement I is incorrect and Statement II is correct.

(4) Both Statement I and Statement II are correct

38) Elevation in boiling point was 0.52 °C when 6 g of a compound x was dissolved in 100 g of water. Molecular weight of x is : ($K = 0.52$ K Kg mol^{-1})

(1) 120

(2) 60

(3) 100

(4) 342

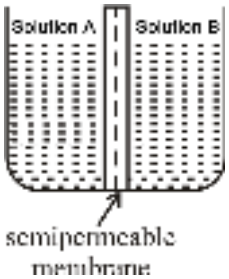
39) Molarity of pure liquid HCl if density of solution is 1.17 g/cc. :

(1) 36.5

(2) 18.25

(3) 32.05

(4) 42.10



40) In the above setup solvent flows from A to B then :

(1) Solution A is more dilute then solution B

(2) Solution A is more concentrated then solution B

(3) Solution A and solution B are equally concentrated

(4) Solution A is at higher temperature relative to solution B

41) At 40°C, the vapour pressure (in torr) of solution containing methyl alcohol and ethyl alcohol is represented by the equation.

$P = 119X_A + 135$ where X_A is mole fraction of methyl alcohol, then the value of $\lim_{x_A \to 1} \frac{P_A}{X_A}$ is

(1) 254 torr

(2) 135 torr

(3) 119 torr

(4) 140 torr

42) The molecular weight of methanoic acid in benzene as determined by depression in freezing point method corresponds to :

(1) Ionization of methanoic acid

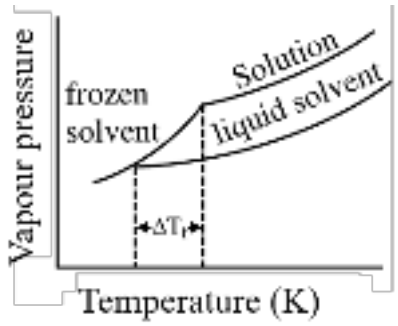
(2) Dimerization of methanoic acid

(3) Trimerization of methanoic acid

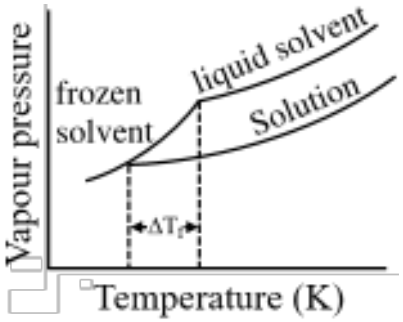
(4) Solvation of methanoic acid

43) Correct graph showing depression of freezing point of solvent when a non-volatile solute is added to it, is-

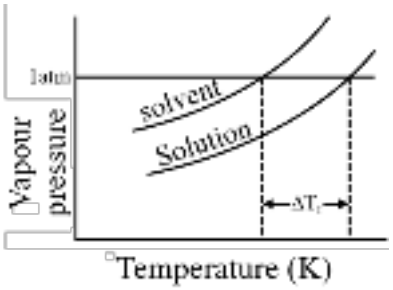
(1)

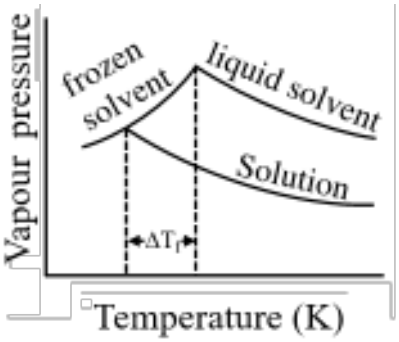


(2)

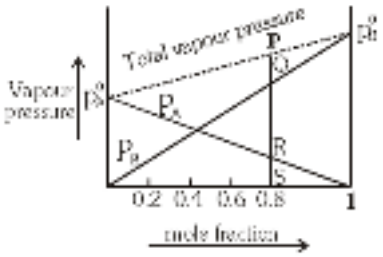


(3)



(4) 

44) Consider the following vapour pressure-composition graph, SP is equal to :-



(1) PQ + RS

(2) QR + RS

(3) SR + SQ

(4) PQ + QR

45) Consider the following solutions:
(a) 0.5 M C_2H_5OH (aq)
(b) 0.1 M $Mg_3(PO_4)_2$ (aq)
(c) 0.25 M NaCl (aq)
(d) 0.125 M Na_3PO_4
Assume all salts to be strong electrolyte then:

(1) 0.5 M C_2H_5OH (aq) highest osmotic pressure

(2) 0.1 M Na_3PO_4 has highest osmotic pressure

(3) 0.125 $Mg_3(PO_4)_3$ has highest osmotic pressure

(4) All have same osmotic pressure.

BIOLOGY

1) Consider following genetic disorders.
Haemophilia, Sickle cell anaemia, Cystic fibrosis, Thalassemia, Phenyl ketonuria, Down syndrome, Klinefelter's syndrome, Turner's syndrome
How many of the above disorders are Mendelian disorders?

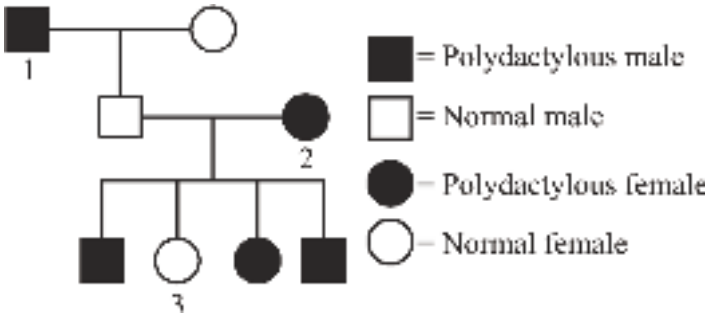
(1) 5

(2) 2

(3) 8

(4) 3

2) In humans, polydactyly (i.e., presence of extra fingers and toes) is determined by a dominant autosomal allele (P) and the normal condition is determined by a recessive allele (p). Find out the possible genotypes of family members 1, 2 and 3 in the given pedigree.



	1	2	3
(1)	PP	Pp	pp
(2)	PP	PP	pp
(3)	Pp	PP	Pp
(4)	Pp	Pp	pp

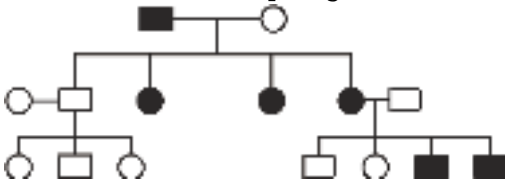
(1) 1

(2) 2

(3) 3

(4) 4

3) Given below is a pedigree of a family suffering from a genetical disorder.



After observing the inheritance pattern, find out that which of the following disorders / conditions can produce such type of pedigree?

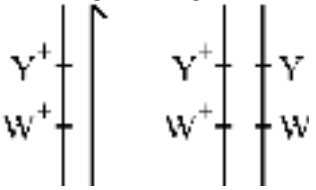
(1) Hypertrichosis

(2) Colourblindness

(3) Haemophilia

(4) Pseudoricketes

4) How many types of gametes will be produced by male & female drosophila respectively having following arrangement of two incompletely linked genes ?



(1) Two & two

(2) Two and four

(3) One & two

(4) Four & four

5) **Assertion :** Egg is responsible for sex of the birds and not the sperm.
Reason : Bird shows female heterogamy.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

6) **Statement-I :** Morgan carried out several dihybrid crosses in *Drosophila* to study genes that were autosomal linked.
Statement-II : Results of Morgan's dihybrid crosses were different to those of dihybrid crosses carried out by Mendel in peas.

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both Statement-I and Statement-II are correct

(4) Both Statement-I and Statement-II are incorrect

7) Find incorrect match from column I and II.

	Column-I	**Column-II**
(1)	Haemophilla	Sex linked recessive
(2)	Sickle cell anaemia	Autosome linked recessive
(3)	Phenyl ketonuria	Autosome linked dominant
(4)	Turner syndrome	Sex chromosomal monosomy

(1) 1

(2) 2

(3) 3

(4) 4

8) In Morgan's experiment, what was the percentage of recombination in case of body colour and eye colour ?

(1) 37.2%

(2) 1.3%

(3) 98.7%

(4) 27.3%

9) Male homogamety can be seen in

(1) Human beings
(2) *Drosophila*
(3) Fowls
(4) Honey bees

10) In Drosophila, four genes shows A-B-C-D arrangement & have following recombination frequencies :
A & B → 2.8%
A & C → 7.8%
D & B → 8.0%
D & C → 3.0%
What would be the distance between genes B & C ?

(1) 3.8 cM
(2) 3 cM
(3) 5.0 cM
(4) 10 cM

11) How many types of gametes will be produced by a trihybrid individual, in which two genes are completely linked ?

(1) Eight
(2) Four
(3) Six
(4) Two

12) How many linkage group are present in female Drosophila ?

(1) 8
(2) 4
(3) 5
(4) 3

13) Morgan and his group found that when genes were grouped on the same chromosome, some genes were very tightly linked and shown:

(1) Very low recombination
(2) Very high recombination
(3) 0% recombination
(4) 100% parental combination

14) Gene silencing using RNAi technique has been applied to make :

(1) Nematode resistant plants
(2) Edible vaccines

(3) Iron fortified rice
(4) Vitamin enriched cereals

15) Which of the following peptide chains is/are not present in mature insulin?

(1) A-peptide
(2) B-peptide
(3) C-peptide
(4) A & B peptides

16) A transgenic food crop which may help in solving the problem of acquired night blindness in developing countries is:

(1) Bt-brinzal
(2) Bt-Soyabean
(3) Golden rice
(4) Flavr-savr tomatoes

17) The microbes providing a natural system for gene transfer to produce transgenic plants is :

(1) *E.coli*
(2) *Pseudomonas putida*
(3) *Bacillus thuringinesis*
(4) *Agrobacterium tumefacines*

18) CryIIAb endotoxins obtained from *Bacillus thuringensis* are effective against :

(1) Nematodes
(2) Cotton bollworms
(3) Mosquitoes
(4) Corn borers

19) Elution is a method applied for :

(1) Making the matrix during gel electrophoresis
(2) Staining the bands of DNA after electrophoresis
(3) Cutting out the piece of agarose gel and extraction of DNA from gel piece
(4) Joining the specific DNA with the cloning vector

20) How many of the following restriction endonuclease enzymes form sticky ends of DNA fragments?
ECoRI, Hind III, Bam I, Hae III, ECoRV

(1) Three
(2) Four
(3) One

(4) Two

21) Which reagent is used to precipitate DNA from solution (aqueous phase) during the process of DNA isolation ?

(1) Phenol

(2) Ehtidium bromide

(3) CCl_4

(4) Chilled ethanol

22) How many of the following statements are correct ?
(a) Ligation of desired gene in coding sequence of enzyme β– galactosidase results into the production of blue coloured colonies in presence of chromogenic substance.
(b) EcoRI cuts the DNA strand at pallindromic sequence and produce blunt ends.
(c) Cellulase, chitinase are the enzymes used to break bacterial cell and to release DNA along with other macromolecules.
(d) In mammals, insulin is synthesized as a fully matured and functional hormone and consist of a single peptide chain.

(1) zero

(2) 1

(3) 2

(4) 3

23) If we want to ligate a foreign DNA at amp^R gene of vector p^{BR322}, then it should be cut by :

(1) BamH I

(2) Sal I

(3) Pvu I

(4) Pvu II

24)

Match the Phylum in Column-I with its characteristic Symmetry in Column-II:

Column-I		**Column-II**	
(a)	Sponges	(i)	Radial Symmetry
(b)	Coelenterates	(ii)	Bilateral Symmetry
(c)	Annelids	(iii)	Mostly Asymmetrical
(d)	Echinoderms (Adult)		

(1) a-(iii), b-(i), c-(ii), d-(i)

(2) a-(i), b-(ii), c-(iii), d-(i)

(3) a-(iii), b-(ii), c-(i), d-(ii)

(4) a-(ii), b-(iii), c-(i), d-(i)

25) **Assertion (A) :-** Members of Platyhelminthes are called acoelomates.
Reason (R) :- In Platyhelminthes, the body cavity is not lined by mesoderm; instead, mesoderm is present as scattered pouches.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

26) Read the statements regarding Porifera:
(I) They have a water transport system where water enters through osculum and exits through ostia.
(II) The body is supported by a skeleton made up of spicules or spongin fibres.
(III) Fertilisation is internal and development is indirect.
Choose the correct option:

(1) I and II are correct

(2) II and III are correct

(3) I and III are correct

(4) All are correct

27) Match the common name with the scientific name:

Common Name		Scientific Name	
(a)	Portuguese man-of-war	(i)	*Meandrina*
(b)	Sea-pen	(ii)	*Physalia*
(c)	Sea-fan	(iii)	*Pennatula*
(d)	Brain coral	(iv)	*Gorgonia*

(1) a-(ii), b-(iii), c-(iv), d-(i)

(2) a-(ii), b-(iv), c-(iii), d-(i)

(3) a-(i), b-(iii), c-(iv), d-(ii)

(4) a-(iv), b-(ii), c-(i), d-(iii)

28) Which distinctive feature is primarily used for locomotion in Ctenophores like *Pleurobrachia*?

(1) Tentacles with cnidoblasts

(2) Eight external rows of ciliated comb plates

(3) Eight internal rows of ciliated comb plates

(4) Jointed appendages

29) **Statement-I :-** Specialised cells called flame cells help in osmoregulation and excretion in flatworms.

Statement-II :- They are mostly endoparasites found in animals including human beings.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

30) Which of the following is NOT a feature of Aschelminthes?

(1) Body is circular in cross-section.
(2) Alimentary canal is incomplete.
(3) They are pseudocoelomates.
(4) Excretory tube removes body wastes through excretory pore.

31) Select the correct match for Annelida:

(1) *Nereis* - Monoecious
(2) *Hirudinaria* – Dioecious
(3) *Pheretima* - Closed circulatory system
(4) Neural system - Single ventral nerve cord

32) Complete the table:

Feature	Description
Body Division	Head, Thorax, (A)
Excretion	(B)
Balance Organs	(C)

(1) A-Abdomen, B-Nephridia, C-Statocysts
(2) A-Tail, B-Malpighian tubules, C-Radula
(3) A-Abdomen, B-Malpighian tubules, C-Statocysts
(4) A-Trunk, B-Flame cells, C-Tentacles

33) **Assertion (A) :-** In Molluscs, the body is unsegmented with a distinct head, muscular foot, and visceral hump.
Reason (R) :- A soft and spongy layer of skin forms a mantle over the visceral hump.

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

34) The water vascular system in Echinoderms helps in all of the following EXCEPT:

(1) Locomotion
(2) Capture and transport of food

(3) Respiration

(4) Excretion of nitrogenous waste via kidneys

35) Identify the correct statement regarding Hemichordata:

(1) They have a rudimentary structure in the collar region called stomochord.

(2) They are mostly fresh-water animals.

(3) Fertilisation is internal.

(4) Circulatory system is of closed type.

36)

In Urochordata, the notochord is present _____, whereas in Cephalochordata, it extends _____.

(1) throughout life; only in larval tail

(2) only in larval tail; from head to tail region and is persistent throughout life

(3) from head to tail; only in larval tail

(4) only in larval tail; from head to tail but disappears in adults

37) **Statement-I:-** Cyclostomes are marine but migrate for spawning to fresh water.
Statement-II:- Their larvae, after metamorphosis, return to the ocean.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

38) Match the feature in Column I with the Class in Column II:

Column-I		**Column-II**	
(a)	Placoid scales	(i)	Osteichthyes
(b)	Operculum present	(ii)	Chondrichthyes
(c)	Air bladder present		
(d)	Ventral mouth		

(1) a-(ii), b-(i), c-(i), d-(ii)

(2) a-(i), b-(ii), c-(ii), d-(i)

(3) a-(ii), b-(ii), c-(i), d-(i)

(4) a-(i), b-(i), c-(ii), d-(ii)

39) **Assertion (A) :-** Osteichthyes fishes do not have to swim constantly to avoid sinking.
Reason (R) :- They possess an air bladder which regulates buoyancy.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

40) Given below are two statements regarding the reproduction and development in specific phyla:
Statement-I :- In Porifera, fertilization is internal and development is indirect having a larval stage which is morphologically distinct from the adult.
Statement-II :- In Ctenophora, sexes are not separate, reproduction takes place only by sexual means, and fertilization is external with indirect development.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

41) Select the option that correctly matches the Organism (Column I) with its Scale Type (Column II) and Distinctive Physiological Feature (Column III):

	Column-I (Organism)	**Column-II (Scale Type)**	**Column-III (Feature)**
(1)	*Scoliodon* (Dog fish)	Placoid scales	Air bladder absent; swims constantly
(2)	*Labeo* (Rohu)	Placoid scales	Air bladder present; regulates buoyancy
(3)	*Trygon* (Sting ray)	Cycloid scales	Possesses poison sting; Operculum present
(4)	*Clarias* (Magur)	Ctenoid scales	6-15 pairs of gill slits; Jawless

(1) 1

(2) 2

(3) 3

(4) 4

42) Which among the following is correct development sequence?

(1) Pollen mother cell → Pollen grain → Microspore tetrad → Male gametes

(2) Sporogenous tissue → Pollen mother cell → Microspore tetrad → Pollen grain → Male gametes

(3) Sporogenous tissue → Pollen mother cell → Pollen grain → Microspore tetrad → Male gametes

(4) Microspore tetrad → Microspore → Pollen mother cell → Male gametes

43) The pistil of Papaver is:

(1) Monocarpellary

(2) Multicarpellary syncarpous

(3) Multicarpellary apocarpus

(4) Absent

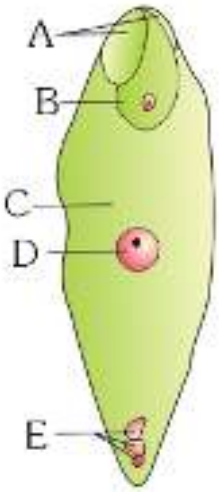
44) Parthenocarpy can be induced through the application of __A__ and such fruits are __B__ .

(1) A - Growth hormone, B - Seeded

(2) A - Growth hormone, B - Seedless

(3) A - Fertilization, B - Seeded

(4) None of these



45) The given diagram is of the fertilised embryo sac:

Statement-1: D is triploid and B is diploid.

Statement-2: E is degenerating synergid.

(1) Both statements are correct

(2) Statement-1 is correct and statement-2 is incorrect

(3) Statement-1 is incorrect and statement-2 is correct

(4) Both statements are incorrect

46) Read the following statements carefully :-

(i) Egg apparatus consist of two synergids and antipodal cells

(ii) Transfer of pollen grains from the anther to the stigma of another flower of the same plant is called geitonogamy

(iii) Pollen grains coming in contact with stigma is a chance factor in both wind and water pollination.

(iv) Pollination by water is more common

How many above statements are correct

(1) One

(2) Two

(3) Three

(4) Four

47) Read the statements carefully :-
(i) Flowers are morphological and embryological marvel and the site of sexual reproduction.
(ii) The proximal end of the filament is attached to the thalamus or the petal of the flower.
(iii) The number and length of stamens are variable in flowers of different species.
(iv) A typical anther is bilobed and tetrasporangiate.
How many above statements are correct :-

(1) One

(2) Two

(3) Three

(4) Four

48) In a typical stamen the proximal end of the filament is :-

(1) Attached to the thalamus

(2) Generally bilobed structure

(3) Towards the stigma

(4) Attached to anther

49) The **correct** sequence of embryo development in most of the Angiosperms is:

(1) Torpedo → Heart shaped → Globular

(2) Heart shaped → Globular → Torpedo

(3) Globular → Heart shaped → Torpedo/Mature

(4) Globular → Torpedo → Heart shaped

50) Multicarpellary apocarpous gynoecium is found in :-

(1) *Hibiscus, Papaver*

(2) *Michelia*, Lotus

(3) *Michelia, Papaver*

(4) *Papaver*, Rose

51) A typical female gametophyte of angiosperm consists of :-

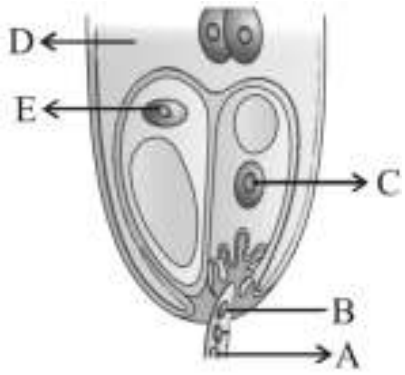
(1) Egg cell, synergids and antipodal cells only.

(2) Egg apparatus, antipodal cells and central cell only.

(3) Egg apparatus and antipodal cells only.

(4) Antipodal cells and synergid cell only.

52) Identify the plants parts labelled A, B, C, D and E in the figure given below and select the right



option.

	A	B	C	D	E
(1)	Male gametes	Vegetative nucleus	Central cell	Egg cell	Synergid cell
(2)	Generative nucleus	Male gamete	Central cell	Synergid cell	Egg cell
(3)	Male gametes	Vegetative nucleus	Synergid cell	Central cell	Egg cell
(4)	Vegetative nucleus	Male gamete	Synergid cell	Central cell	Egg cell

(1) 1

(2) 2

(3) 3

(4) 4

53) The viability of pollen grains of wheat is :-

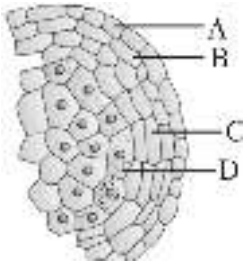
(1) 30 hours

(2) 30 seconds

(3) 30 days

(4) 30 minutes

54) Given below is diagramatic sketch of a transverse section of young anther identify the parts



labelled A, B, C and D and select the right option about them:-

	Part A	Part B	Part C	Part D
(1)	Epidermis	Middle layer	Endothecium	Tapetum
(2)	Epidermis	Endothecium	Tapetum	Middle layer
(3)	Epidermis	Endothecium	Middle layer	Tapetum

(4)	Epidermis	Endodermis	Middle layer	Sporogenous tissue

(1) 1

(2) 2

(3) 3

(4) 4

55) Which of the following plant came in India as contaminant with imported wheat and causes pollen allergy :-

(1) Chenopodium

(2) Sorghum

(3) Parthenium

(4) Castor

56) Number of spiracles present in *Periplaneta americana* is :

(1) 10

(2) 9

(3) 18

(4) 20

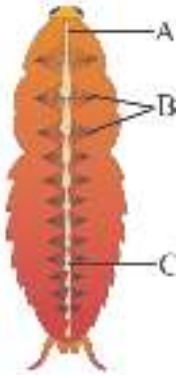
57) In cockroach nerve cord ganglia are

(1) 2 pair thoracic and 4 pair abdominal

(2) 3 pair thoracic and 6 pair abdominal

(3) 3 pair thoracic and 5 pair abdominal

(4) 2 pair thoracic and 7 pair abdominal



58) See the diagram carefully and select the correct match :

(1)	(A)	Anterior Aorta	Help in excretion
(2)	(B)	Alary muscles	Help in blood circulation
(3)	(C)	Alary muscles	Help in excretion
(4)	(B)	Pulsatile Ampulla	Help in blood formation

(1) 1

(2) 2

(3) 3

(4) 4

59) Select the incorrect statement regarding the *Periplaneta americana* ?

(1) Body is about 34-53 cm long
(2) Body is divided in head, thorax and abdomen
(3) Each body segment is covered by sclerites
(4) Body consists of 20 embryonic segment

60) Which of the following structure is responsible for sexual dimorphism in male and female cockroach ?

(1) Anal styles
(2) Head
(3) Anal cerci
(4) Wings

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	3	3	3	3	3	4	1	1	2	4	2	1	2	4	4	3	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	1	3	1	3	4	2	3	1	4	2	3	1	2	4	4	2	2	3
Q.	41	42	43	44	45															
A.	2	3	2	3	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	3	2	4	3	4	1	1	4	3	3	2	2	2	1	2	4	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	4	1	2	4	1	3	1	2	4	3	4	3	1	2	4	2	3	1
Q.	86	87	88	89	90															
A.	1	2	2	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	4	4	2	1	2	3	2	3	3	2	2	1	1	3	3	4	2	3	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	3	1	3	2	1	2	1	2	3	3	2	4	1	2	1	1	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	2	2	2	2	4	1	3	2	2	4	4	3	3	4	2	2	1	1

SOLUTIONS

PHYSICS

1) Note that drift velocity in semiconductors is affected by an increase in temperature in the same way as in metals i.e., reduced due to an increase in the number of collision with the lattice ions. which vibrate with greater amplitude at a higher temperature.

2)

$\sigma = ne(\mu_e + \mu_r)$

$\sigma = 2 \times 10^{19} \times 1.6 \times 10^{-19} (0.39 + 0.14)$

$= 1.6\ (\Omega - m)^{-1}$

$$R = \rho\frac{\ell}{A} = \frac{\ell}{\sigma A} = \frac{0.5 \times 10^{-3}}{1.6 \times 10^{-4}} = \frac{25}{8}\Omega$$

$$i = \frac{V}{R} = \frac{2 \times 8}{25} = 0.64A$$

3) Cut in voltage of diode $V_C = 0.4$ V

$$I = \frac{V - V_C}{R}$$

(Forward resistance is very small in comparison to R)

$$I = \frac{10 - 0.4}{12000}$$

I = 0.8 mA

4) (A) Peak current in diode

$$= \frac{20 - 0.7}{500 + 10} = 37.8 \text{ mA}$$

(B) Peak voltage across the load

$= (37.8 \text{ mA}) \times 500\ \Omega = 18.9$ V

(C) Peak current (if diode is ideal)

$$= \frac{20}{500} = 40 \text{ mA}$$

(D) Peak voltage across the load (diode ideal)

$= 40 \text{ mA} \times 500 = 20$ V

5)

In solar cell, I-V characteristics are obtained in fourth quadrant.

6) Apply distributive law

(1) $B + A\bar{B} = (B + A) \cdot (B + \bar{B})$ (using distributive law)

$= (B + A). 1 = B + A$

(2) $A \bullet (\bar{A} + B) = A \bullet \bar{A} + A \bullet B = 0 + AB = AB$

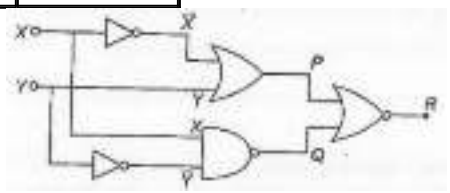
(3) $A + B \bullet C = (A + B) \bullet (A + C)$

(4) $(\bar{A} + B) \bullet (A + C) = \bar{A} \bullet A + \bar{A} \bullet C + B \bullet A + B \bullet C = \bar{A}C + BA + BC$

7)

The truth table can be written as

X	Y	$\overline{X}$	$\overline{X}$	$P = \overline{X} + Y$	$Q = \overline{X + Y}$	$R = \overline{P + Q}$
0	1	1	0	1	1	0
1	1	0	0	1	1	0
1	0	0	1	0	0	1
0	0	1	1	1	1	0



So, X = 1, Y = 0 gives output R = 1 However, based strictly on the truth table, option 3 correct.

Alternate method :

$R = \overline{(\bar{X} + Y)} + (X \cdot \bar{Y})$

$\Rightarrow R = (\bar{X} + Y) \cdot (X \cdot \bar{Y})$

$\Rightarrow R = (X \cdot \bar{Y}) \cdot (X \cdot \bar{Y})$

$R = (X \cdot \bar{Y})$

For R = 1, X = 1, Y = 0

8)

$\lambda = 600$ nm

Energy of photon

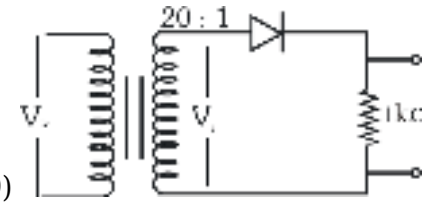
$E = \frac{1240}{\lambda(nm)} eV$

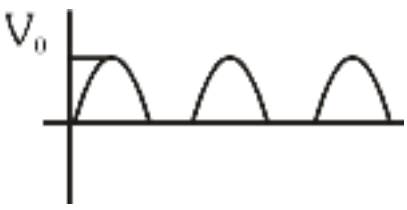
$= \frac{1240}{600} = 2.06$ eV

$E < E_g$

So, not detect the light

9)



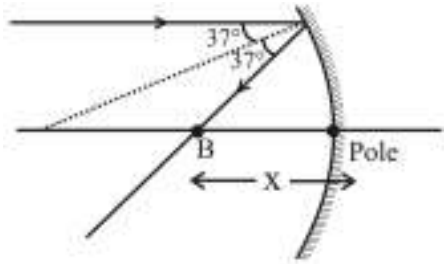


$\frac{n_p}{n_s} = \frac{V_p}{V_s}$

$\Rightarrow V_s = \frac{n_s}{n_p} \times V_p = \frac{1}{20} \times 220 = 11$ V(peak)

Here $V^0 = 11$ V so $V_{RMS} = \frac{V_0}{2} = \frac{11}{2} = 5.5$ V $I_{RMS} = \frac{5.5}{10k} = 0.55$ mA

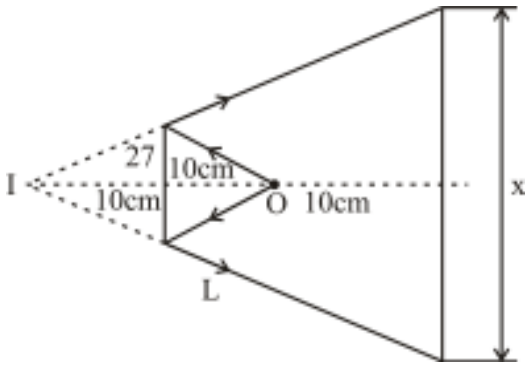
10)



The distance of point B from pole is given by

$$|x| = R - \frac{R}{2\cos i}$$

$$= R - \frac{R}{2\cos 37^\circ} = \frac{3R}{8} \quad (\cos 37^\circ = 4/5)$$



11)

From two similar triangles in the diagram we can write.

$$\frac{x}{27} = \frac{30}{10}$$

x = 81cm

Image is visible for a distance on AB = 81cm

Now using equation of motion for calculating time.

$$S = \frac{1}{2}at^2$$

$$81 = \frac{1}{2} \times 2 \times t^2$$

$$t = \sqrt{81} = 9\,\text{sec}$$

12)

For real object :

(A)	m = –4	Image is real, inverted and enlarged it is possible in concave mirror only.

(B)	$m = -\frac{1}{4}$	Image is real, inverted and small in size it is possible in concave mirror only.
(C)	m=+3	Image is virtual, errect and enlarged it is possible in concave mirror only.
(D)	$m = +\frac{1}{3}$	Image is virtual, errect and diminished it is possible only in convex mirror.

13) u = –40

m = –4

$-4 = \frac{f}{f+40}$

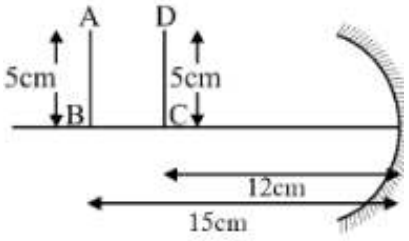
$f + 40 = -\frac{f}{4}$

$\frac{5f}{4} = -40$

f = –32

So R = 64 cm

14)



For AB

$\frac{1}{-15} + \frac{1}{V_1} = -\frac{1}{10}$

$\Rightarrow V_1 = -30cm$

For CD

$\frac{1}{-12} + \frac{1}{V_2} = -\frac{1}{10}$

$\Rightarrow V_2 = -60cm$

Length of Image wire

$= AB\left|\frac{V_1}{U_1}\right| + CD\left|\frac{V_2}{U_2}\right| + |V_2 - V_1|$

$= 5 \times \frac{30}{15} + 5 \times \frac{60}{12} + (60-30)$

= 10 + 25 + 30

= 65 cm

15)

u = –45 cm f = –30 cm

$\vec{V}_I - \vec{V}_m = -m^2(\vec{V}_0 - \vec{V}_m)$

$V_I - 2 = -\left(\frac{f}{f-u}\right)^2 (20-2)$

$V_I = 2-\left(\frac{-30}{-30+45}\right)^2 \times 18$

$\Rightarrow V_I = 2 - 72 = -70$ cm/s

16) Lateral shift = $\frac{t\sin(i-r)}{\cos r}$

Here $1.\sin 60° = \sqrt{3}\sin r$

$\Rightarrow$ $r = 30°$, $t = 5\sqrt{3}$cm

Lateral shift = $5\sqrt{3}\frac{\sin(60°-30°)}{\cos 30°} = 5$cm

17) $F = \sqrt{x_1 x_2}$

$10 = \sqrt{5 \times x_2} \Rightarrow x_2 = 20$

from lens = 20 + 10 = 30 cm

18) $\vec{P}_{net} = 15D - 5D = 10D$

$\Rightarrow P = \frac{100}{F} \Rightarrow F_{net} = \frac{10}{P_{net}} = \frac{100}{10}$

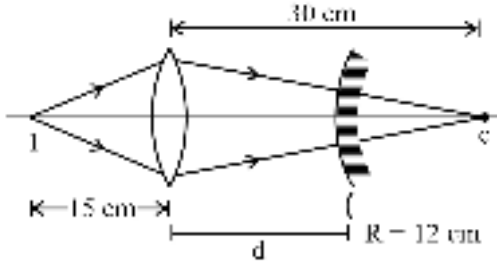
$F_{net} = 10$ cm

Now $\frac{1}{v} - \frac{1}{-20} = \frac{1}{10} \Rightarrow v = 20$ cm

19)

u = –15 cm, f = 10

$\frac{1}{v} - \frac{1}{-15} = \frac{1}{10}, \frac{1}{v} = \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30}, v = 30$ cm



d = 30 – 12 = 18 cm

20) $P = P_1 + P_2 \Rightarrow 10 = \frac{1}{f_1} + \frac{1}{f_2}$

$P = P_1 + P_2 - dP_1P_2$

$6.25 = \frac{1}{f_1} + \frac{1}{f_2} - \frac{0.20}{f_1 f_2}$

On solving (3) option is obtained.

$P_1 = 2.5$ D

$P_2 = 7.5$ D

21)

A simple microscope uses a convex lens to magnify objects. When the object is placed between the optical centre and the focal point of the convex lens, the lens forms a virtual, erect, and magnified image of the object. This principle is fundamental in understanding the functioning of magnifying glasses and simple microscopes.
Remember that the simple microscope is essentially a convex lens used as a magnifier - the key is object placement between focal point and optical centre for magnification.

22)

$m = m_0 \times m_e$

$$m = \left[\frac{f_0}{f_0 + u_0}\right] \times \frac{D}{f_e}$$

$$m = \left[\frac{4}{4} - 4.5\right] \times \frac{24}{8}$$

$= 24$

23) for eye piece
$v_e = -25$ cm ; $f_e = 6.25$ cm $u_e = ?$

$$\frac{1}{-25} - \frac{1}{-u_e} = \frac{1}{6.25}$$

$$u_e = \frac{25 \times 6.25}{25 + 6.25} = 5\text{cm}$$

separation between both lenses $L = v_0 + |u_e|$
$15 = v_0 + 5$
$v_0 = 10$ cm
for objective lens
$v_0 = 10$ cm ; $f_0 = 2$ cm, $u_0 = ?$

$$\frac{1}{10} - \frac{1}{u_0} = \frac{1}{2}$$

$$\frac{1}{u_0} = \frac{1}{10} - \frac{1}{2}$$

$$= \frac{1-5}{10} = \frac{-4}{10} = \frac{-2}{5}$$

$u_0 = -2.5$ cm

24)

$$P = 1.25 = \frac{100}{f} \qquad P = -20D = \frac{100}{f}$$

$f = 80$ cm $\qquad f = -5$ cm
$L = f_o - f_e = 80 - 5 = 75$ cm

$$m = \frac{f_o}{f_e} = \frac{80}{5} = 16$$

25)

Magnification, $m = \frac{f}{f-u}$

When u = –5cm, image virtual

$m_1 = +m\ m_1 = \frac{f}{(f-(-5))} = \frac{f}{(f+5)}$

When u = –15 cm, the image is real and $m_2 = -m$

$m_2 = \frac{f}{f-(-15)} = -\frac{f}{f+5}$

$f + 5 = -(f + 15) \Rightarrow 2f = -15 - 5 = -20$

$\Rightarrow f = -10$ cm

26) Apparent depth of the object = $\frac{\text{Real depth}}{\text{Refractive index}} = \frac{33.25}{1.33}\text{cm}$

Distance of object from the mirror = $15 + \frac{33.25}{1.33} = 40$ cm

Similarly, distance of image from the mirror = $15 + \frac{25}{1.33} = 33.8$ cm

For concave mirror, u = –40 cm and ν = –33.8 cm

Applying mirror formula, $f = \frac{u\nu}{u+\nu} = \frac{(-40)(-33.8)}{(-40-33.8)} = -18.3\text{cm}$

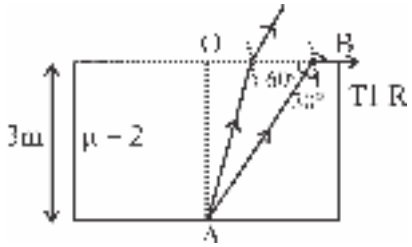
27) $\frac{f_0}{f_e} = 20,\ f_0 + f_e = 2.1$

$f_0 = 200$cm, $f_e = 10$cm, final magnification = $f_0\left[\frac{1}{D} + \frac{1}{f_e}\right] = 28$

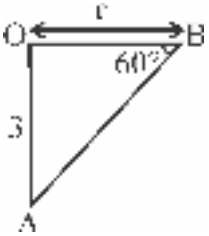
28) The correct figure will show: 1. A ray of white light incident on the air-water interface. 2. Multiple refracted rays (at least three distinct colors should be shown: red, green, and blue are common choices). 3. The refracted rays should be *bent towards the normal* (the line perpendicular to the surface) because light is going from a rarer medium (air) to a denser medium (water).

4. The refracted rays should be *fanning out*, with violet bending the most and red bending the least.

29)



ΔOAB



$$r = \frac{3}{\tan 60}$$
$r = \sqrt{3}$
Area = π(3) = 3π

30)

$$\frac{1}{f} = \left(\frac{\mu_\ell}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$\frac{1}{2} = (1 \cdot 5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \Rightarrow 1 = \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$\frac{1}{f'} = \left(\frac{1 \cdot 5}{1 \cdot 25} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$\frac{1}{f'} = \left(\frac{0.25}{1 \cdot 25}\right) \times 1 \Rightarrow f' = 5 \text{ cm}$$

31)

$L = f_0 + f_e$
$10 = f_0 + f_e$ .....(1)
$M = \frac{f_0}{f_e} \Rightarrow 4 = \frac{f_0}{f_e}$
$\Rightarrow f_0 = 4f_e$ put in (1)
$10 = 4f_e + f_e \Rightarrow f_e = 2cm$
So, $f_0 = 10 - 2 = 8$ cm
So, lens $L_4 = 8$ cm objective
$L_1 = 2$cm eye piece.

32) $$M = \frac{v_0}{u_0}\left(1 + \frac{D}{f_e}\right) = m_0\left(1 + \frac{D}{f_e}\right)$$
$$30 = m_0\left(1 + \frac{25}{5}\right) \Rightarrow m^0 = 5$$

33) As, critical angle, $$i_C = \sin^{-1}\left(\frac{\mu_r}{\mu_d}\right)$$
where, n_r and n_d in the refractive index of rarer and denser medium, respectively
For total internal reflection at surface AB, $\theta > i_C$

$$\Rightarrow \quad \theta > \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) \qquad \text{.....(i)}$$

Also, for total internal reflection at surface CD,

$$\theta > \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) \qquad \text{.....(ii)}$$

For both Eqs. (i) and (ii) to be correct, $$\theta > \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) > \sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$

$\Rightarrow \theta > \sin^{-1}(\sin 60°) \Rightarrow \theta > 60°$

$\therefore \theta_{min} = 60°$

34)

Polarizing angle = $\tan\theta = \mu \Rightarrow \sqrt{3} = \mu$

Critical angle $\sin i_C = \frac{1}{\mu}$

$\Rightarrow i_C = \sin^{-1}\left(\frac{1}{\mu}\right) = \sin^{-1}(0.577)$

35) **Explain the Question :** To find the graph between transmitted intensity and the angle between analyser and polariser.

Concept : This question is based on Malus law.

Solution :

$I = I_0 \cos^2\theta$

I_0 : Intensity of plane polarised light

I : Intensity of transmitted light

θ : Angle between transmission axes of polariser and analyser

$\theta = 0°$, $I = I_0$

$\theta = 90°$, $I = 0$

Final Answer : (2)

36) **Explain the Question :** To find the fraction of intensity of incident light that passes through the system.

Concept : This question is based on Malus law.

Solution :

$I = I_0\cos^2\phi$, $\phi = 60°$

$\xrightarrow{(I_0)} P_1 \xrightarrow{\left(\frac{I_0}{2}\right)} P_2 \xrightarrow{\left(\frac{I_0}{8}\right)} P_3 \xrightarrow{\left(\frac{I_0}{32}\right)} P_4 \xrightarrow{\left(\frac{I_0}{128}\right)} P_5 \xrightarrow{\left(\frac{I_0}{512}\right)}$

Final Answer : (4)

37)

$\beta = \frac{\lambda D}{d}$

$\beta' = \frac{\lambda(2D)}{d/2} = 4\beta$

38) distance of 4th bright fringe

$y = 4\beta = \frac{4\lambda D}{d}$

$\lambda = \frac{yd}{4D} = \frac{1.2 \times 10^{-2} \times 0.28 \times 10^{-3}}{4 \times 1.4}$

$\lambda = 600$ nm

39) $x = \frac{(\mu - 1)tD}{d}$

4th Dark location = $\frac{(2 \times 4 - 1)D\lambda}{2d}$

$\frac{(\mu - 12)tD}{d} = \frac{7D\lambda}{2d}$

$t = \frac{7\lambda}{2(\mu - 1)}$

$= \frac{7 \times 6 \times 10^{-7}}{2(1.5 - 1)}$

$= 4.2\ \mu m$

40)

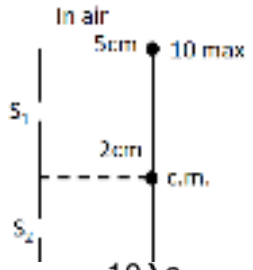
Explain Question:

To find the y-coordinates for central and 10th maximum if Y.D.S.E is immersed in liquid of $\mu = 1 \cdot 5$

Concept:

Y D S E in medium.

Solution:



$10\beta_a = \frac{10\lambda a}{d} = (5 - 2)\ cm.$

$10\beta_a = 3cm$

$\beta_a = \frac{3}{10} cm$

$\beta_a = \frac{\beta_a}{\mu} cm$

$\beta_a = \frac{\frac{3}{10} cm}{3/2} = \frac{2}{10} = \frac{1}{5} cm = 0 \cdot 2cm$

Position of maximum in liquid

$10\beta_m = 0 \cdot 2cm \times 10 = 2cm$

y coordinate of $10^{th} Max = 2 + 2$

Position = 4cm

Central maximum will not shift, so its position remains as 2cm

$\therefore (2cm, 4cm)$

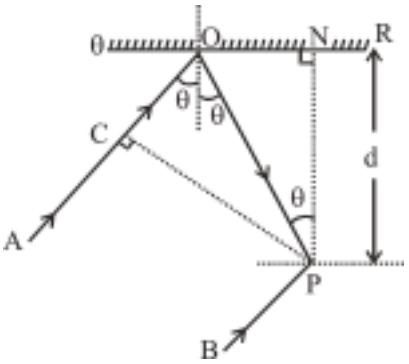
(Central maximum,10th maximum)

Final Answer: Option (3)

41)

In ΔPNO

$\cos\theta = \frac{d}{OP}$



OP = dsec θ

ln ΔPCO

$\cos 2\theta = \frac{OC}{OP}$

OC = dsecθ × cos2θ

At point P,

OP + OC + λ/2 = nλ [n = 1]

(dsecθ) + (dsecθ × cos2θ) = λ – λ/2

dsecθ (1 + cos2θ) = λ/2

$\cos\theta = \frac{\lambda}{4d}$

42) Part of wave front will take shape of mirror

43) $y^{n^{th}}_{B(R)} = y^{(n+1)^{th}}_{B(r)}$

$\frac{n\lambda_R D}{d} = \frac{(n+1)\lambda_B D}{d}$

$\Rightarrow n\lambda_R = (n+1)\lambda_B$

$\Rightarrow n(7800) = (n+1)(5200)$

$\Rightarrow 6n = 4(n+1) \Rightarrow 2n = 4$

$\Rightarrow n = 2$

44)

Intensity at the centre will be zero if path difference = $\frac{\lambda}{2}$

or $(\mu-1)t = \frac{\lambda}{2}$ or $t = \frac{\lambda}{2(\mu-1)}$

45) Given that, $\frac{I_1}{I_2} = 2x$

We know,

$I_{max} = \left(\sqrt{I_1}+\sqrt{I_2}\right)^2$ & $I_{min} = \left(\sqrt{I_1}-\sqrt{I_2}\right)^2$

$$\square \quad \frac{I_{max}-I_{min}}{I_{max}+I_{min}} = \frac{2\sqrt{I_1 I_2}}{I_1+I_2} = \frac{2\sqrt{I_1/I_2}}{\frac{I_1}{I_2}+1} = \frac{2\sqrt{2x}}{2x+1}$$

CHEMISTRY

46)

		Salt	Nature	
(a)	K_2S	SBWA	Basic	pH > 7
(b)	NaCl	SASB	Neutral	pH = 7
(c)	NH_4Cl	WBSA	Acidic	pH < 7
(d)	Salt of weak acid & weak base	WBWA	can be acidic or basic	

47)

Fact Based

48) Conjugate base of H_2O is OH^-
Conjugate base of HF is F^-

49) (B) $H_2PO_4^- \rightarrow HPO_4^{2-} + H^+$
acid C.B.
(C) $H_2O \rightleftharpoons H^+ + OH^-$
Temp ↑ Kw ↑

50)

$$CH_3COOH \xrightarrow{NaOH} CH_3COONa$$

Initial 1 mole 0
1/4 th 3/4 mole 1/4 mol →pH_1
3/4 th 1/4 mole 3/4 mole→pH_2
$pH_1 = pK_a + \log 1/3$
$pH_2 = pK_a + \log 3$
$\therefore \Delta pH = 2\log 3$

51)

due to common ion solubility decreases.

52) $\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{C_2}{C_1}} \Rightarrow \frac{0.01}{\alpha_2} = \sqrt{\frac{0.4}{0.1}}$

$\alpha_2 = 0.005$ or % $\alpha_2 = 0.5$

53)

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

C – Cα		Cα	Cα + 0.1
≃ C		Cα	≃ 0.1 M

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$$1.8 \times 10^{-5} = \frac{[C\alpha][0.1]}{[0.2]}$$

54) $NH_4KHPO_3 \rightleftharpoons \overset{\oplus}{N}H_4 + \overset{\oplus}{K} + HPO_3^{\ominus}$

$K_{sp} = S^3$ and $S = (K_{sp})^{1/3}$

55)

For Acidic Solution :
$[H^+] > [OH^-]$
$P^H < 7$
$P^{OH} > 7$
$[H^+] > 10^{-7}$
$[OH^-] < 10^{-7}$
Formula:
$[H^+] > [OH^-]$
$P^{OH} > 7$
$[OH^-] > 10^{-7}$
Correct Ans: Option: 4
All of these.

56)

Air is mixture of gases.

57) E and V

58)

$w = - P_{ext} (V_2 - V_1)$
$= - 1 \times (50 - 15)$
$= - 35$ L-atm

59)

$\Delta H = \Delta U + \Delta ngRT$

$$-1366.5 = \Delta U - \frac{1 \times 8.314 \times 300}{1000}$$

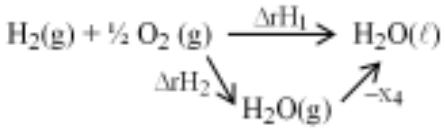
$\Delta U = - 1364$ KJ/mol

60)

$H_2O(l) \quad \rightleftharpoons \quad H_2O\,(g)$
1 atm, 298 K 1 atm, 298 K
This process is non spontaneous at this condition. So $\Delta G < 0$.

61)

Explain Question : Match thermodynamics process in column-I with their corresponding relation ship in column-II
Concept : This Question is based on $\Delta E = q + w$
Solution :
i → Adiabatic process ; $q = 0$
$\Delta E = w$ (b)
i - (b)
ii. Isochoric process ; $w = 0$
$\Delta E = q$ (c)
ii - (c)
iii. Isothermal process ; $\Delta E = 0$
$q = - w$ (d)
iii - (d)
iv - Isobaric process ; $\Delta H = q$ (a)
iv - (a)
i - (b), ii - (c), iii - (d), iv- (a)
Final Answer option (1)



62)
$\Delta rH_2 = E_{H-H} + 1/2\ E_{O=O} - 2E_{O-H}$
$= x_1 + 1/2 \times x_2 - 2x_3$
$\Delta rH_2 + x_4 = \Delta rH_1$
$\Rightarrow \Delta rH_1 = x_1 + 1/2 \times x_2 - 2x_3 - x_4$

63)

(a) $N_2(g) + O_2(g) \rightarrow 2NO(g)$;

$\Delta H = X_1 = 2\Delta H_f NO(g)$

$\Delta H_f\ NO(g) = \frac{\Delta H}{2}$

64)

 Heat released for hydrogenation of 1 mol double bond = x

 Heat released for hydrogenation of 2 mol double bond in  < 2x

Because some energy will be consumed to break resonance.

65) $P(s) + \frac{3}{2}Cl_2(g) \rightarrow PCl_3(g)$

$306 = 314 + \frac{3}{2} \times 2 \times 121 - 3E_{P-Cl}$

$3E_{P-Cl} = 371$

$E_{P-Cl} = 123.66$ kJ/mol

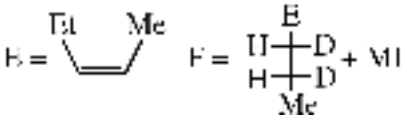
66)

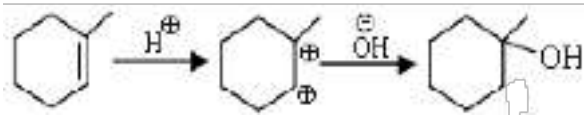
EAR

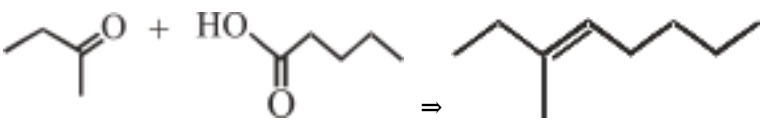
67)

A = $CH \equiv C^{\ominus}$ B = $CH \equiv C - CH_3$

C = $C^{\ominus} - Me$ D = $Et - C \equiv C - Me$



68) 

69) 

70)

ESR

71)

ESR

72)

NCERT XI Pg. # 382

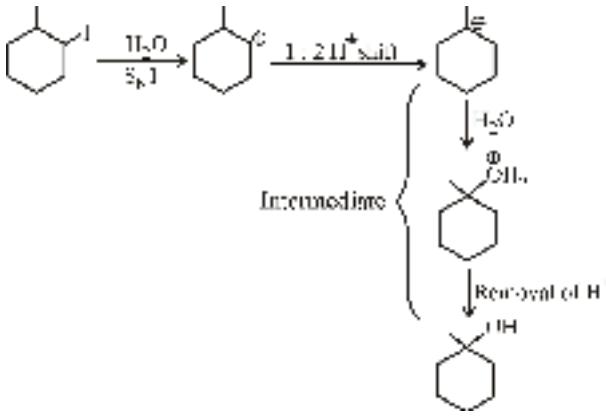
73)

x = S_N1 Reaction (it is hydrolysis)
y = E_2 ($RO^{\ominus}$ is strong base causes elimination)

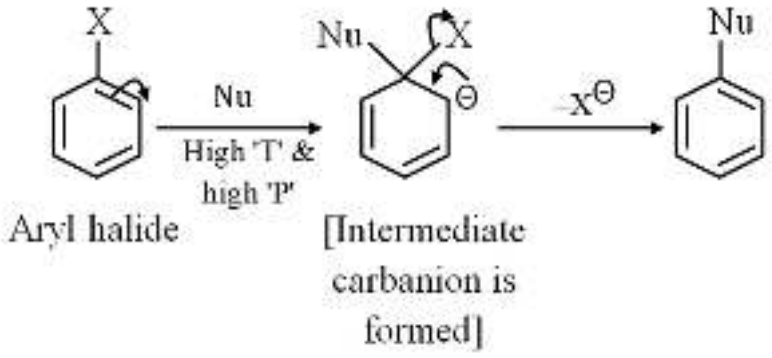
74) Reaction 1 : SN_1
Reaction 2 : E_2
SN_1 is independent of concentration of nucleophile/base

75) Allyl bromide (i) is most reactive among the given halides as pi bonds from alylic position stabilizes the transition state. Electron withdrawing inductive effect of CH_3O^- increase reactivity of (iv) over (iii).

76)



77)



78)

Due to more stable carbocation

79)

Because $(CH_3)_3\ CO^{\ominus}\ K^{\oplus}$ is bulky base

80) Nucleophilicity ∝ e^- donating tendency
$CH_3CH_2–S^{\ominus}$ > $CH_3–O^{\ominus}$> Ph–$CH_2–O^{\ominus}$> $CH_3–O–CH_2–O^{\ominus}$

81)

A. Ideal solution: A–A, B–B, and A–B interactions are identical; $\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$.
B. Positive deviation: A–B interactions < A–A, B–B → $\Delta H_{mix} > 0$, $\Delta V_{mix} > 0$.
C. Negative deviation: A–B interactions > A–A, B–B → $\Delta H_{mix} < 0$, $\Delta V_{mix} < 0$.

Solution:

A. Benzene + Toluene → Ideal → Q
B. Ethanol + Water → Positive deviation → P, R ($\Delta H_{mix} > 0$)
C. Benzene + Ethanol → Positive deviation → P, R ($\Delta H_{mix} > 0$)
D. Acetone + Chloroform → Negative deviation → P, S ($\Delta V_{mix} < 0$)

82)

$P = K_H X$ ⇒ mole fraction is unit less.
 unit of K_H = unit of pressure

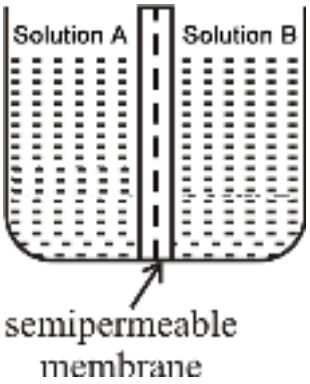
83)

Elevation in boiling point (ΔTb) = 0.52 °C
Mass of solute (w) = 6 g
Mass of solvent (W) = 100 g = 0.1 kg
Ebullioscopic constant (Kb) = 0.52 K kg mol^{-1}
Using the formula $\Delta T_b = K_b \times w/(M \times W)$, we get $0.52 = 0.52 \times 6/(M \times 0.1)$.
Solving for M gives M=60

84) $$M = \frac{\rho \times 1000}{\text{Molar mass}} = \frac{1.17 \times 1000}{36.5} = 32.05$$

85)

Solvent flows from A to B, solution A must have a lower solute concentration (be more dilute) to drive osmosis towards the more concentrated solution B.

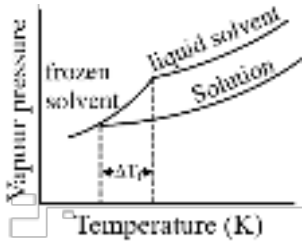


Final Answer – Option (1)

86) $P=119X_A + 135$;

$$\lim_{x_A \to 1} \frac{P_A}{X_A} = 119 + 135 = 254 \text{ torr}$$

87) NCERT XII Pg. # 56 # Part-I
All the molecules of methanoic acid undergo association (dimerise) in benzene and hence the molar mass will be calculated on the basis of ΔT_f.

88) 

89)

$P_S = P_A^0 \cdot x_A + P_B^0 \cdot x_B$
$SP = P_A^0 \times 0.2 + 0.8 \times P_B^0$
SP = SR + SQ

90)

(a) π = icRT
= 1 × 0.5 RT = 0.5 RT
(b) π = icRT
= 5 × 0.1 RT = 0.5 RT
(c) π = icRT
= 2 × 0.25 RT = 0.5 RT
(d) π = icRT
= 4 × 0.125 RT = 0.5 RT

BIOLOGY

91)

NCERT-XII Pg. # 88

92)

Refer to module

93)

Refer to module

94)

NCERT XII Pg. # 68

95) NCERT Page No. 86, 87 (E), 95 (H)

96) NCERT-XII, Pg. No. # 67

97) NCERT Pg. # 74, 75

98) NCERT (XII) Pg. # 84

99) **Question Explanation :**
The question asks to identify a species from the given options that exhibits female heterogamety.

Concept :
This question is based on sex determination.

Solution :
"Birds have a ZW type of sex determination system where females are heterogametic (ZW) and males are homogametic (ZZ)."

Final Answer :
Option (3)

100) **Asking about:**
The question asks for the distance between genes B and C, given the crossover values (COV) between different gene pairs in Drosophila.

Concept:
Gene Mapping: Crossover values are used to determine the relative distances between genes on a chromosome. 1% crossover frequency is approximately equal to 1 map unit (mu) or 1 centimorgan (CM).
Linear Gene Arrangement: The genes are arranged linearly, meaning they are in a straight line on the chromosome.

Solution:
1. Visualize the Gene Order: The question states the gene order is A-B-C-D.
2. Use Given COV Values:
A & B: 2.8 CM
A & C: 7.8 CM

D & B: 8.0 CM
D & C: 3.0 CM
3. Calculate the Distance Between B and C:
Since the genes are in the order A-B-C-D, we can infer that:
The distance between A and C is the sum of the distances between A and B and B and C.
■Therefore, distance(B-C) = distance(A-C)-distance(A-B)
■distance(B-C) = 7.8 CM-2.8 CM = 5.0 CM
4. Verify with D:
• Similarly, we can verify using the D values:
The distance between D and B is the sum of the distances between D and C and C and B.
■Therefore, distance(B-C) = distance(D-B) - distance(D-C) ■distance(B-C) = 8.0 CM - 3.0 CM = 5.0 CM
Answer:
The distance between genes B and C is 5.0 CM. Therefore, the correct answer is 3.

101)

Refer to module

102)

Refer to module

103)

Refer to module

104)

NCERT XII Pg. # 180

105)

NCERT XII Pg. # 181, 182

106)

NCERT XII Pg. # 179

107)

NCERT XII, Pg # 209

108)

NCERT XII, Pg # 209

109)

NCERT XII Pg. # 168

110)

Refer to module

111)

NCERT XII, Pg # 201

112)

NCERT XII, Pg # 200, 196, 201, 211

113) NCERT Page No. 169(E), 185(H)

114)

NCERT Pg. # 38, 39
Sponges are asymmetrical, Coelenterates & adult Echinoderms are radial, Annelids are bilateral.

115) NCERT Pg. # 39
Reason is false because "mesoderm as scattered pouches" describes pseudocoelomates (Aschelminthes), not acoelomates.

116) NCERT Pg. # 40
Statement I is incorrect because water enters through ostia and exits through the osculum.

117) NCERT Pg. # 41
Physalia is Portuguese man-of-war, Pennatula is Sea-pen, Gorgonia is Sea-fan, Meandrina is Brain coral.

118) NCERT Pg. # 42
Ctenophores use eight external rows of ciliated comb plates for locomotion.

119) NCERT Pg. # 42
Both statements are correct; flame cells aid in excretion and they are mostly endoparasites.

120) NCERT Pg. # 43
Aschelminthes have a complete alimentary canal with a muscular pharynx, not incomplete.

121) NCERT Pg. # 43
Pheretima (Earthworm) has a closed circulatory system; *Nereis* is dioecious.

122) NCERT Pg. # 44
Arthropod body has Head, Thorax, **Abdomen**; Excretion by **Malpighian tubules**; Balance by **Statocysts**.

123) NCERT Pg. # 44
Both facts are true, but the mantle (R) describes the shell-secreting layer, not the reason for the body division (A).

124) NCERT Pg. # 45
Echinoderms lack an excretory system; the water vascular system helps in locomotion, food transport, and respiration.

125) NCERT Pg. # 45
Hemichordates have a stomochord in the collar region; they are marine, not freshwater.

126)

NCERT Pg. # 46
In Urochordata, notochord is in larval tail only; in Cephalochordata, it is head-to-tail and persistent.

127) NCERT Pg. # 47
Both statements are correct; Cyclostomes migrate to fresh water to spawn, and larvae return to the ocean.

128) NCERT Pg. # 47-48
Placoid scales/Ventral mouth = Chondrichthyes; Operculum/Air bladder = Osteichthyes.

129) NCERT Pg. # 48
Correct; Osteichthyes don't need constant swimming because the air bladder provides buoyancy.

130) NCERT Pg. # 40, 42
Porifera has internal fertilization/indirect development; Ctenophora has external fertilization/indirect development.

131)

NCERT Pg. # 48
Scoliodon is a cartilaginous fish with placoid scales and lacks an air bladder.

132) Sporogenous tissue → Pollen mother cell → Microspore tetrad → Pollen grain → Male gametes

133) Multicarpellary syncarpous

134) A - Growth hormone, B - Seedless

135) Statement-1 is correct and statement-2 is incorrect

136) NCERT-XII, Page#27,28,29

137)

NCERT Based

138)

Question Explanation: Question is asking about ,the proximal end of the filament of typical stamen is attached to which structure

Concept: structure of stamen

Solution:

The proximal end of the filament is attached to the thalamus or the petal of the flower.

Final answer: 1

139)

Explain Question: Question is asking about ,the correct sequence of embryo development in most of the Angiosperms

Concept: Post-fertilisation events-Embryo

Solution: In angiosperm the zygote gives rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo/ Torpedo

The correct sequence of embryo development- Globular → Heart shaped → Torpedo/Mature

Final answer: 3

140) NCERT (XI) Pg. # 75
NCERT (XII) Pg # 25

141) NCERT XII, Page # 10,11

142) **Explaining the Question :** Question asking about the labelled figure of embryo sac when entry of pollen tube in synergid. **Concept :** Fertilisation Solution : In given figure labelled parts are A - Vegetative nucleus, B-Male gamete C-Synergid cell, D- Central cell, E-Egg cell

143)

NCERT Based

144) NCERT (XII) Pg. # 22

145) NCERT Pg. # 23(E) & 25(H)

146)

Question Explanation: Number of cockroach spiracles.

Concept: Respiration in cockroach.

Solution: The cockroach, Periplaneta americana, has 10 pairs of respiratory openings called spiracles. This makes a total of 20 spiracles.

Final Answer: 20

147) NCERT-XI, Pg#114

148)

Question Explanation: See the diagram carefully and select the correct match?

Concept: COCKROACH

Solution: OPTION B IS CORRECT MATCH, THAT ALARY MUSCLES HELPS IN BLOOD CIRCULATION

Final answer: 2

149) NCERT Pg # 111 para 7.4.1

150) NCERT Pg # 112 para 7.4.1